



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-014C Release Configuration Snapshot

C3-00 competency Gate | Subscription data release

Artifact	C3_014C_RELEASE_CONFIGURATION_SNAPSHOT_GATE_RC1
Status	PROTECTED FRESH RETEST - DO NOT OPEN UNLESS ASSIGNED
Teaching standard	M01 beginner-first teacher loop; book remains sufficient without video.
Bridge boundary	Course 2B competencies may be assumed; all new engineering concepts still start from first principles.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House rule: understand -> follow once -> see expected result -> break/fix -> do a changed version -> validate -> explain -> save evidence. A running command alone never counts as mastery.

Gate conditions

No lesson book, practice review, web/tutorial answer or AI solving during the independent Gate attempt. You may use normal tool `--help`/official command reference only if the Gate instructions explicitly permit it. Freeze the entire attempt before opening its review.

CASE

You receive the `Release Configuration Snapshot` case bundle containing config.csv, subscriptions.csv, README_CASE.txt. Build/repair a small data utility and operational handoff so another engineer can run and trust it.

REQUIRED DELIVERABLES

1) written contract/grain; 2) Linux-native project evidence; 3) reusable Python core + CLI; 4) config/secret rule; 5) logs + stable failure behavior; 6) tests incl. edge/failure; 7) Git branch/commits + clean ignore; 8) rerun proof; 9) runbook; 10) Docker proof if available or honest approved fallback.

VALIDATION RULE

Reconcile row/file controls to source, validate required schema/business rule independently, and demonstrate one deliberately broken case before declaring success.

PASS RULE - NO SCORE SHORTCUT

Every required competency C1-C10 must be PASS (C9 may be FALLBACK-VERIFIED only when Docker is genuinely unavailable and blocker/design evidence meets the approved fallback). Critical failures must equal zero. There is no percentage, point total or average that can compensate for a failed competency.

COMPETENCIES

ID	Competency	PASS evidence	Status
C1	Engineering contract & grain	Input/output grain, rules, failure, rerun and independent validation are explicit.	PASS / FAIL / N/A*
C2	Linux/filesystem safety	Correct shell/path; inspect before destructive action; source/evidence handling safe.	PASS / FAIL / N/A*
C3	Git change hygiene	Reviewable branch/commits; no secret/venv/bytecode; diff understood.	PASS / FAIL / N/A*
C4	Python environment reproducibility	Clean venv/dependency recreation proven.	PASS / FAIL / N/A*
C5	Structure/config/secret boundary	Core/CLI boundary + documented config precedence + no secret leakage.	PASS / FAIL / N/A*
C6	Logging/failure semantics	Actionable logs; known failure non-zero; unexpected error not swallowed.	PASS / FAIL / N/A*
C7	Testing	Tests protect contract incl. failure/edge case; deliberate regression is catchable.	PASS / FAIL / N/A*
C8	CLI contract	Useful help, stable invocation/output and machine-readable failure behavior.	PASS / FAIL / N/A*
C9	Container reasoning	Image/container/mount/runtime reasoning correct; real Docker evidence or honest approved fallback.	PASS / FAIL / N/A*
C10	Idempotence/reproducibility/run book	Safe rerun demonstrated; operator can reproduce/validate/recover from runbook.	PASS / FAIL / N/A*

*C9 is not truly N/A: use FALLBACK-VERIFIED only under the documented Docker-unavailable path.

CRITICAL FAILURES

- Real secret/token/key exposed in submitted source/log/evidence.
- Fabricated command/runtime/Docker evidence.



- Destructive unsafe operation or data loss without justified recovery control.
- Contract failure exits success or publishes a known-invalid result.
- Protected review opened/copied before first attempt is frozen.
- First-attempt evidence overwritten/rewritten after review.
- Unresolved plausible wrong result with no independent validation.

FREEZE YOUR ATTEMPT

Save under `evidence/GATE\_C/attempt\_001/`; create an attempt manifest/hash or otherwise freeze file list + contents before opening C3-014C review. Append corrections only as a later repair/retest artifact.

INTERVIEW DEFENSE

Prepare 3-5 minutes: architecture, input/output grain, one failure, one test, rerun strategy, why your validation is independent, and one tradeoff you would change at larger scale.